ITM SKILLS UNIVERSITY

Department of Computer Science & Engineering

SOFTWARE ENGINEERING & PROJECT MANAGEMENT

PETPAL

Case Study No. 64

UML PACKAGE

System Analysis & Architectural Design Models (Use Case, Class, Sequence, Activity & State Diagrams)

SUBMITTED BY

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Project: PetPal — Appointment Booking and Records App for a Veterinary Clinic Chain
Case Study: No. 64
Subject: Software Engineering & Project Management

DIAGRAM 1 & 2

1. Use Case Diagram

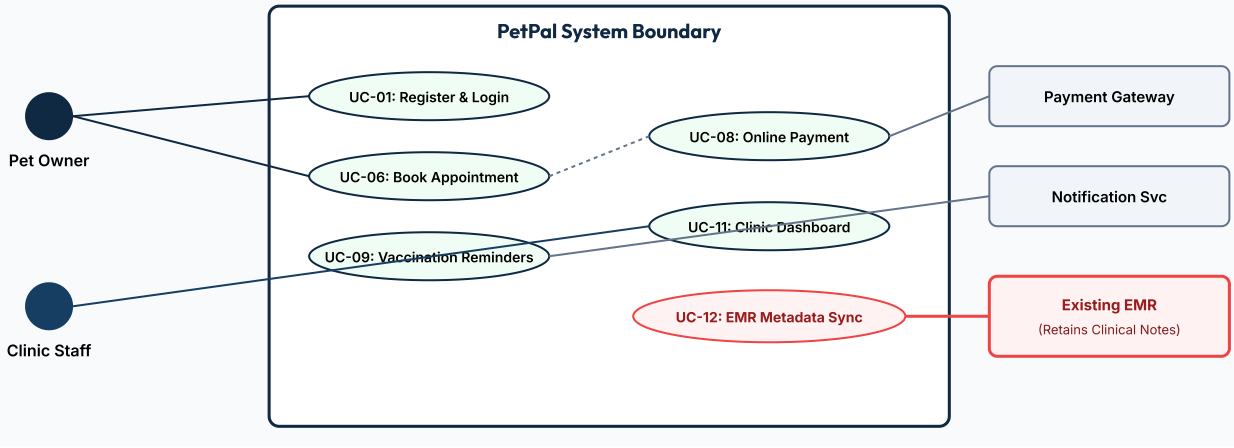


Figure 1.1: PetPal System Use Case Diagram

2. Domain Class Diagram (Zero ClinicalNote Entity)

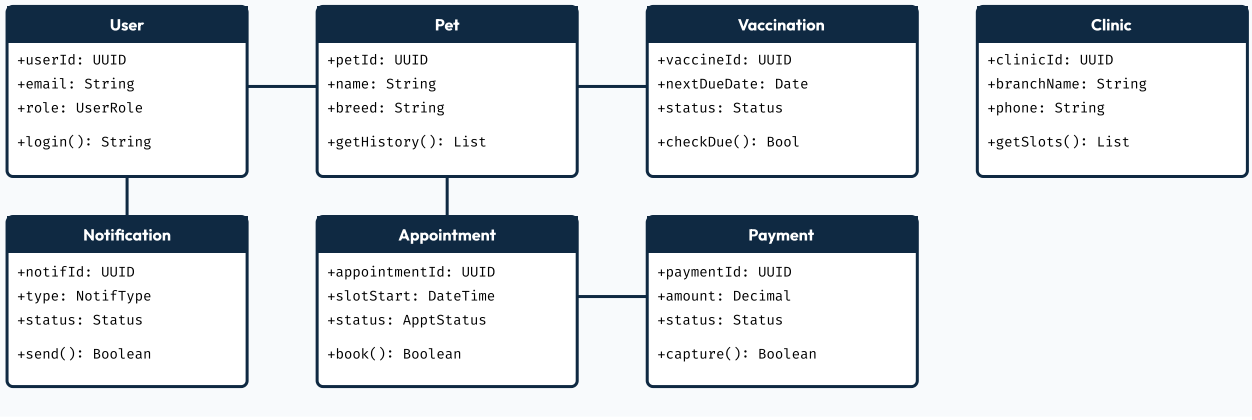


Figure 2.1: Domain Class Diagram (Zero ClinicalNote Entity)

DIAGRAM 3, 4, 5 & 6

3. Dynamic UML Models (Sequence, Activity & State)

Sequence Diagram 1 (Appointment Booking): Models the 10-minute pessimistic slot reservation hold, payment gateway webhook callback, parallel notification dispatch, and metadata synchronization with the legacy clinic software.

Sequence Diagram 2 (Vaccination Reminders): Models the automated daily 00:01 AM background daemon checking for T+14, T+3, T+1 day booster due dates, checking user reminder preferences, and dispatching SMS/Email alerts.

Activity Diagram: Models the user booking journey: Login → Pet Selection → Clinic Selection → 30-Min Slot Availability Gateway → Temporary 10-Min Hold → Payment Authorization & Retry Logic → Confirmed Booking.

Appointment State Machine: Models all legal state transitions: `Requested → Pending Payment → Confirmed → Checked-in → In Consultation → Completed` (with alternative `Payment Failed` and `Cancelled` states).